Electricity anomaly detection: where the project stands

AT82.01 Computer Programming for Data Science and AI, Dr. Chantri Polprasert
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Written to be read by a teammate who wants to know exactly what was done, how, and what is still open. Section 10 lists what we need to decide together.

In short. We flag the hours when a building uses electricity in a way that is unusual for that building at that time, and we check the flags against stretches that people marked as unusual. The midterm part is done. Since then we reran everything from a clean start, added a second kind of detector (Isolation Forest), added honest margins of error, tested the professor's question about long-lasting problems, reviewed flags that matched no label, and tested on a third dataset (CU-BEMS) with problems we planted ourselves. Main findings: (1) the previous hour is the most useful clue; (2) a very simple rule, "this hour will equal the last hour", is as good as our best learning model at finding labeled events when both flag the same number of hours; (3) the learning model (Model C) is clearly better on long problems, the simple rule on one-hour spikes; (4) the Isolation Forest does not beat either. What remains is mostly the write-up, three small extras, and a few checks between the two of us.

1. The question and the idea

Research question. How well do flags from the gap between actual and expected electricity use match labeled anomalies in building data, and which design choices (features, cut-off, detector) improve that match? The professor added: if the model predicts better from historical use, does it start to overlook anomalies that last a long time, because it learns them as normal?

How the tool works, in plain words. For every building and every hour it works out how much electricity the building should use (from the time, the weather, building facts, and the readings just before). It compares that with the real reading. The gap is divided by that building's own usual gap, which gives a score. An hour whose score is above a cut-off is flagged. We then compare the flags with the labeled stretches.

What a flag means. Unusual for its context. It is never proof of waste, a fault or a saving. The output is a ranked list for a person to check.

2. The three datasets

Dataset	What it is	How we use it
GEPIII	Hourly electricity of about 1,400 buildings in 2016 (ASHRAE Great Energy Predictor III). We use meter 0 (electricity) of the 200 buildings that have labels: 1,749,494 rows. Site 0 is in kBTU, converted to kWh (x 0.2931). Six weather variables per site. Building facts (year built is 66% blank, floor count 76% blank).	Main data. Train, validate and test in time order.
LEAD 1.0	Labels: people looked at graphs of GEPIII buildings and marked unusual hours. 37,296 labeled hours, 5,811 events. In our test period 587 events are usable (631 in validation).	The answer key. Labeled hours are left out when the scaling and cut-offs are set, and (in the retrained version, section 5.6) out of model training.
CU-BEMS	One 7-storey academic office building at Chulalongkorn University, Bangkok. One-minute power in kW, 1 July 2018 to 31 December 2019, 14 files (one per floor per year). No weather, no labels.	Third dataset. We plant problems ourselves, so the right answer is known. Not evidence about real-world performance.

BDG2 (Building Data Genome Project 2) is the source collection that GEPIII comes from, so it is not an independent fourth dataset. The midterm wording "compare with BDG2" is being changed to say this.

3. Method in detail

3.1 Cleaning and set-up

- Complete hourly grid for every building (8,784 hours in 2016). Lags (previous hour, same hour yesterday, same hour last week) are looked up by time, not by row position. The row-based version was wrong for 0.89% of rows.
- Runs of exact zero lasting 24 hours or more are treated as missing (a broken meter, not a quiet building). 70,157 rows (4.01%) are exact zeros. No negative readings, no exact duplicates. No readings are filled in.
- **Split by time:** train before 13 September 2016, validation 13 September to 7 November, test 7 November to 31 December. The test period was already looked at for the midterm; see the open decision in section 10.

3.2 The six predictors

1. **Same as last hour** (prediction = the previous reading).
2. **Historical typical value** (median by building, hour and weekend; falls back to building and hour, then building).
3. **Model A**: gradient boosting (scikit-learn HistGradientBoostingRegressor, on log(1+reading)) with 15 features: calendar, building facts, six weather variables. No history.
4. **Model B**: A plus the same hour yesterday and last week (17 features).
5. **Model C**: B plus the previous hour (18 features).
6. **Fixed percentile**: flags readings above the 99.5th percentile of clean training use.

Model settings: 600 rounds, learning rate 0.05, 127 leaves, at least 100 rows per leaf, l2 1.0, seed 0, no tuning.

3.3 Score, cut-off, flag

- Gap e = log(1+actual) minus log(1+predicted). Score = |e - m| / s, where m and s are the building's own median and scaled median absolute deviation (x 1.4826, floor 0.05) from clean validation hours. Four buildings with fewer than 500 clean validation hours borrow the pooled values.
- One shared cut-off per method: the 0.5% upper quantile of clean validation scores. Hours that LEAD marks as unusual are left out when the scaling and cut-offs are set. The cut-offs are saved before the test period is scored.
- **Isolation Forest** (new): a detector that needs no prediction. One forest per building (100 trees, 256 hours per tree, seed 0), fitted on training hours only, no labels. Variant 1 uses the Model C gap, hour and weekend flag. Variant 2 uses the log reading, the change from the previous hour, the change from 24 hours earlier, hour and weekend flag.

3.4 How we judge a flag

- An event is **found** if at least one of its hours is flagged (the most generous definition). We also report the share of an event's hours that are flagged and the number of hours before the first flag.
- A **run** is consecutive flagged hours. Run precision = share of runs that touch a labeled event.
- **Equal alarm rates.** Because methods flag very different numbers of hours at their own cut-offs, we also compare them when each flags the same share of hours (0.5%, 1%, 2%), either with one threshold for all buildings (pooled) or each building flagging its own top hours (per building).
- **Margins of error** come from resampling whole buildings 2,000 times (seed 0), because events inside one building are related.

4. What was done, in order

Stage	What	Notebook / files
Midterm (before 2 Nov)	Foundations, cleaning, labels joined, six predictors, scores and cut-offs frozen before the test, evaluation, equal-rate comparison, report and slides.	Notebooks 01 to 07
2 Nov: full check	Every notebook run from a clean folder in order. All outputs identical to an earlier rerun, down to the last decimal (deterministic). Wording and number fixes in the older documents.	All notebooks
2 Nov: Isolation Forest, equal rates, margins of error (T24 to T28)	Added the detector; equal-rate comparison with intervals from resampling buildings; check whether choosing cut-offs from labels helps.	Notebook 08; `results/eval_equal_rate_ci_ext.csv`, `eval_frozen_rules_test.csv`
2 Nov: labeled hours out of training (T40)	Retrained Models A, B, C without hours LEAD marks as unusual.	`notebooks/label_excluded/`, `results/label_excluded/`
2 Nov: persistence test (T27)	Recall by event length with intervals, and flag rate by position inside an event.	Notebook 09; `results/persistence_*.csv`
2 Nov: flags that match no label (T29)	Sampled 30 unmatched flagged runs, drew them, rated them (Claude did the rating), tagged all 714 unmatched runs.	Notebook 10; `unmatched_flags_review.pdf`, `unmatched_flags_review_sheet.csv`
9 Nov: CU-BEMS (T31, T32)	Hourly series for 7 floors plus a six-floor total; anomalies planted in the test period; all methods run; 20 random placements.	Notebooks 11 and 12; `results/cu_bems/`

5. Results on GEPIII and LEAD

5.1 Predicting is not detecting (test period)

Predictor	MAE kWh	R2
Same as last hour	8.09	0.993
Historical typical value	30.28	0.958
Model A (no history)	32.70	0.955
Model B (+ yesterday, last week)	19.59	0.978
Model C (+ previous hour)	8.92	0.993

The previous hour is by far the most helpful clue. The best predictor by accuracy (last hour) still finds fewer long events than Model C.

5.2 At the cut-offs set in advance (midterm)

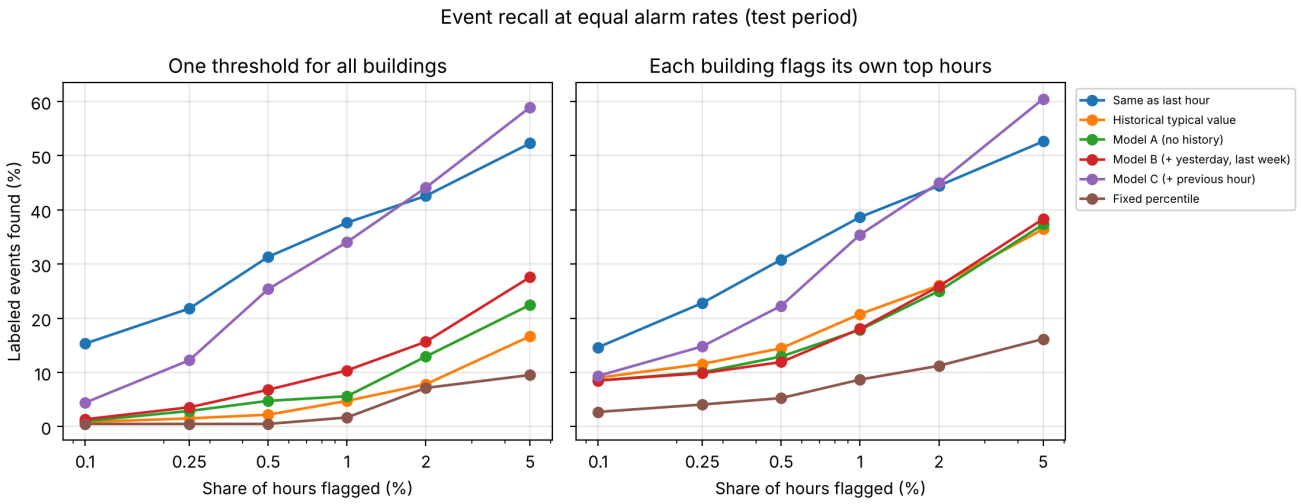
Method	Events found (of 587)	Run precision	Share of hours flagged
Same as last hour	171 (29.1%)	18.5%	0.35%
Historical typical value	8 (1.4%)	14.0%	0.08%
Model A	8 (1.4%)	21.5%	0.08%
Model B	20 (3.4%)	23.7%	0.11%
Model C	171 (29.1%)	21.6%	0.50%
Fixed percentile	42 (7.2%)	3.8%	1.87%

A problem we found and reported. The share of hours flagged at these cut-offs ranged from 0.075% to 1.87% against a 0.5% target. We saw this only after looking at the test results, so we kept the cut-offs, said so, and added the equal-alarm-rate comparison below.

5.3 At equal alarm rates, each building flagging its own top hours (587 events, 95% interval from resampling buildings)

Method	0.5% flagged	1% flagged	2% flagged
Same as last hour	30.8 (25.7 to 36.2)	38.7 (32.8 to 44.5)	44.5 (38.3 to 50.6)
Historical typical value	14.5	20.8	26.1
Model A	12.9	17.9	25.0
Model B	11.9	18.1	25.9
Model C	22.3 (17.9 to 27.2)	35.4 (30.0 to 41.1)	45.0 (39.5 to 50.7)
Isolation Forest (on model C gap)	20.8	28.6	39.2
Isolation Forest (raw readings)	22.3	29.8 (23.9 to 36.5)	41.1
Fixed percentile	5.3	8.7	11.2

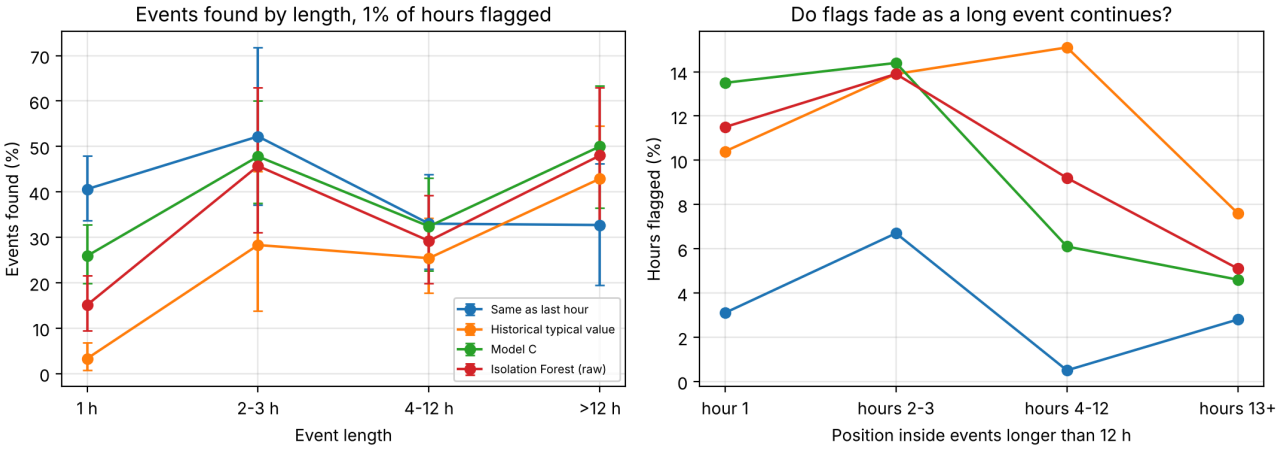
Values are percent of the 587 usable test events found. Model C minus the last-hour rule, with each building flagging its own top hours: -8.5 points at 0.5% (interval -12.8 to -4.5, the last-hour rule is clearly ahead), -3.2 at 1% (-7.7 to +1.0) and +0.5 at 2% (-3.4 to +4.7), so at 1% and 2% the two are a statistical tie. With one shared threshold the differences are -6.0, -3.6 and +1.5 points, and the intervals include zero at 1% and 2%. Model C beats the typical-value rule by 7 to 36 points depending on the setting. The Isolation Forest is below the last-hour rule in most settings.



5.4 The professor's question: long-lasting problems (1% flagged per building, test period)

Event length	Events	Last hour	Model C	C minus last hour (95% interval)
1 hour	212	40.6	25.9	-14.6 (-21.3 to -8.9)
2 to 3 hours	92	52.2	47.8	-4.3 (-19.6 to +7.7)
4 to 12 hours	185	33.0	32.4	-0.5 (-7.3 to +5.7)
over 12 hours	98	32.7	50.0	+17.3 (+7.3 to +29.6)

Answer so far. Yes for the plain copy of the previous hour: after the first hour, the previous hour already contains the problem, so it stops flagging. Much less for Model C, which also looks at yesterday and last week: on events longer than 12 hours it finds about 50 in 100 against about 33 in 100, and that gap is larger than the margin of error. On one-hour spikes it is the other way round. Inside long events every method flags fewer hours as the event goes on (Model C: 13.5% of hours at the start, 4.6% after 12 hours); we do not know why and will say so.



5.5 Does choosing cut-offs from labeled events help? (T26)

We compared the single shared cut-off with a separate cut-off for each building. The per-building cut-offs find more events but only because they flag more hours: for Model C aiming at 1%, the shared cut-off found 39.5% of events while flagging 1.52% of test hours, the per-building cut-offs found 50.8% while flagging 2.53%. For the typical-value rule the per-building cut-off flagged 9.7%

of all hours. Since the gain is mostly a larger alarm budget, and the equal-rate comparison already uses per-building thresholds, we **kept the single shared cut-off** as the main design. (Result file: `results/eval_frozen_rules_test.csv`.)

5.6 Leaving labeled hours out of training (T40)

Our first design left LEAD-marked hours out of the scaling and cut-offs but not out of the models' training data. We retrained Models A, B and C without them. At 1% flagged per building, Model C finds 38.0% of events (it was 35.4%), tied with the last-hour rule (38.7%). The earlier gap to the last-hour rule was therefore partly an artefact of training on the unusual hours. This is a reason to state the result as "a tie" and not as "the simple rule is better".

5.7 Flags that match no label (T29)

Model C at the pre-set cut-off has 714 flagged runs on the test period that touch no labeled event. We sampled 30 (the 15 with the highest peak score plus 15 at random, seed 0), plotted each with the surrounding week, and rated them. Ratings: 18 look really unusual, 4 plausible, 6 false alarm, 2 cannot tell. **Claude made these ratings from the plots**; they are not independent human judgments. The two of us should each rate a few and compare. If they hold, a good part of the low precision comes from stretches nobody labeled, not only from false alarms. This does not prove it. (`results/unmatched_flags_review.pdf` and `unmatched_flags_review_sheet.csv`.)

6. The third dataset: CU-BEMS with planted problems (T31 and T32)

6.1 Why, and its role

The course needs at least three datasets. GEPIII and LEAD have real but imperfect labels. CU-BEMS has no labels, so it cannot measure agreement with people. What it can do is test what the labels cannot: whether each method finds a problem of known size and length, and what happens in a long one. It is an **evaluation dataset**, not a pure test set: the models are trained on CU-BEMS's own earlier months (the models need each building's routine), and problems are planted only in the last three months, which nothing was tuned on. Training on GEPIII and applying to CU-BEMS would not work because CU-BEMS has no weather or building facts.

6.2 How

- 14 files; only power columns (kW) used. Average over the minutes of each hour, sum over the meters on a floor. A meter counts for an hour only with 45 or more of 60 minutes; a floor-hour is valid only if every meter counts; invalid hours stay missing.
- Series: 7 floors plus a six-floor total (floors 1 to 5 and 7). Valid hours: floor 1 96.1%, floor 2 96.1%, floor 3 86.0%, floor 4 76.2%, floor 5 92.6%, floor 6 86.9%, floor 7 95.9%, six-floor total 65.1%. No negative values and no exact zeros.
- Split: train 1 July 2018 to 30 June 2019, validation July to September 2019, test October to December 2019.
- Models A', B', C' (calendar and series number; B' adds yesterday and last week; C' adds the previous hour), the two baselines and one Isolation Forest per series. Same settings as the midterm.
- Planting (test only): for each series 16 problems, every length (1, 3, 8, 24 hours) with every shape (use x1.3, x1.6, x2.0, or dropped to 5% of the reading). They do not overlap, are at least 48 hours apart and lie on real readings. About 119 problems (about 7% of valid test

hours) per placement. History features are recomputed after planting. 20 random placements (seeds 1000 to 1019); we report the mean.

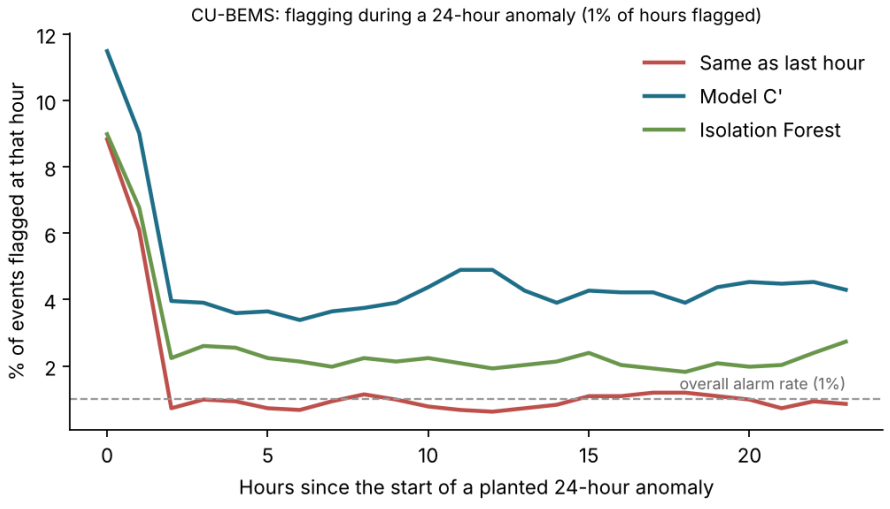
Check needed. The 2019 file for floor 6 (`2019Floor6.csv`) stops on 22 October 2019 at 13:27 and ends with a stray "2" (the beginning of a timestamp), so it looks cut off. Floor 6 therefore has data for only 23% of the test hours. Arpan or Barnnita should compare the file size with the figshare page. If the original is complete we will re-download and rerun (about 20 minutes).

6.3 Results (percent of planted problems found, mean of 20 placements)

Method	1% flagged	2% flagged	5% flagged	Frozen cut-off (hours flagged)
Same as last hour	25	32	44	15 (0.52%)
Historical typical value	9	18	34	3 (0.94%)
Model A'	10	18	35	5 (1.03%)
Model B'	12	18	33	4 (0.80%)
Model C'	27	41	65	24 (0.94%)
Isolation Forest	23	33	47	18 (0.61%)

Model C' minus the last-hour rule: +2.2 points at 1% (but between -5.1 and +8.4 across placements, so not clear), +9.1 at 2% (between +2.5 and +16.0), +21.2 at 5%. By size at 2% flagged (percent found): a drop to near zero is easy (C' 86, last hour 59); a 30% rise is hard for everyone (C' 10, last hour 19, Isolation Forest 15); a doubling is found best by C' (50, against 28 and 36). By length at 2%: C' finds 30, 34, 41 and 59 for 1, 3, 8 and 24 hours; the last-hour rule 20, 25, 31 and 50.

Long problems. For the 24-hour problems the average flag rate in the first 6 hours, the middle hours and the last 6 hours (at 1% flagged): last-hour rule 3.5%, 0.9%, 1.0%; Model C' 6.3%, 4.1%, 4.5%; Isolation Forest 4.7%, 2.1%, 2.2%; Models A', B' and typical value stay level near 5 to 6% (they do not use the previous hour, but they are weak at small rises). So the last-hour rule loses its grip on a long problem, and Model C' loses less. It agrees with the LEAD result.



Limits to state beside every CU-BEMS number: one building and its floors are related; the problems are simple and our own design; there are no human labels; only recall is measured and flags outside the planted problems are not necessarily false alarms; Thai public holidays are not modelled (not checked); no weather; floor 6 is short; nothing

was tuned on CU-BEMS. Differences from the earlier plan: problems planted in the test period only; 16 per series (about 7% of hours instead of 5%); 5% alarm rate added.

7. What we can honestly say

1. Flags from an expected-use model find a limited share of labeled events (about 35 to 45 in 100 when 1 to 2% of hours are flagged), and most flagged runs do not overlap a label. Some of that may be real unusual use nobody labeled (our small sample suggests so; to be cross-checked).
2. The previous hour is the most useful clue. The very simple last-hour rule ties with Model C at equal alarm rates and is better on one-hour spikes.
3. Model C is better on long events (over 12 hours: about 50 against 33 in 100) and, on CU-BEMS planted problems, at 2% flagged and above.
4. Better prediction from history does not hide long anomalies for Model C, but copying the last hour does.
5. The Isolation Forest is not better than either.
6. Choosing cut-offs per building from labels did not help once the alarm budget is held equal.

We will not claim: that a flag is waste, a fault or a saving; that the system beats a simple baseline; that CU-BEMS results show real-world performance.

8. Known limits

- Labels are human judgments from graphs, and LEAD comes from GEPIII, so they are not independent evidence. We measure agreement with people, not truth.
- 200 labeled buildings, one year, one split. Models are tested on a later period for the same buildings, not on new buildings.
- "Found" needs only one flagged hour, the most generous definition. Small groups (36 events in the Education-only view) make differences of a few points noise.
- The test period was seen for the midterm. We made every new choice on validation only and score the test period once per final design.
- Public data has no occupancy, schedules or maintenance records, so causes cannot be named.

9. What is left (deadline 17 November)

Task	What	Size
T38	Simple filter: mark a flag as explained when most buildings of a site deviate together in the same hour; count removed flags and check recall.	0.5 day
T30	Ranked review list: top flagged runs per building with score, date, actual and expected use; one-page guide.	0.5 day
T39	Which features matter (permutation importance) for interpretability.	0.5 day
T33	Final tables and figures from the result files.	1 day
T44 to T47	Map the report to the AIT template; Related works with verified citations; Discussions (performance, interpretability, complexity); honest comparison with a baseline.	3 to 4 days
T34, T35, T37	Report, slides, script and Q&A, presentation.	3 to 4 days
T36	Tidy the folder: delete `CP-LAB/_rerun_tmp`, final packaging (`requirements.txt` is written).	0.5 day
Optional	T23, T41 to T43 (more history clues and similar); dropped first if time runs short.	-

10. What we need to decide or do together

1. **Test protection.** We already looked at the whole test period for the midterm. Current rule (Option A): every new choice on validation only, test scored once per final design, and the report says the test period was also used for the midterm. This keeps all 587 events. Alternative (Option B): split the test period in two; cleaner but about half the events. Do you agree with A?
2. **Cross-check the unmatched-flag ratings.** Each of us rates about 10 of the 30 runs in `unmatched_flags_review.pdf` without looking at Claude's notes, then compare.
3. **Floor 6 file.** Compare `2019Floor6.csv` with figshare.
4. **Holiday filter.** The midterm report promised a filter for known causes such as holidays. GEPIII has no holiday calendar per site. Either build a weather-only/site-wide version (T38) or drop the holiday wording.
5. **Division of the write-up.** Proposal: one of us takes Introduction, Problem statement, Related works, Datasets; the other Methodology, Results, Discussions, Conclusions; both read each other's part and the slides.
6. **CU-BEMS framing.** Are we comfortable calling it an evaluation dataset trained on its own earlier months?

11. Where everything is

Item	Location (project folder CP-LAB/Project)
Notebooks in run order	`notebooks/01 ... 12` and `notebooks/label_excluded/`; `notebooks/README.md` explains each. Run 01 to 06, then 08, 09, 10, then 11 and 12, then 07.
Results	`results/`, `results/label_excluded/`, `results/cu_bems/`
Data	`data/cu_bems/` (raw), `data/cu_bems_hourly/` (hourly series); large GEPIII files stay out of the repository
Packages	`requirements.txt` (Python 3.13, numpy 2.5.3, pandas 3.0.5, scikit-learn 1.9.1, matplotlib 3.11.2)
Documents	Project Master Document (everything, in detail), the Plain English overview, the Complete Task List (47 tasks: 31 done, 1 partly done, 15 to do) and `docs/` (decision.md, project_explained.md, ACTIONPOINTS.md)

12. Words used

Anomaly / unusual hour: use that is unusual given its context. **Residual / gap:** actual minus expected use. **Score:** the gap divided by that building's usual gap. **Cut-off:** the score above which an hour is flagged. **Alarm rate:** the share of hours flagged. **Recall (events found):** share of labeled events with at least one flagged hour. **Run precision:** share of flagged runs that touch a labeled event. **Isolation Forest:** a detector that finds hours that are easy to separate from the rest. **Lag:** an earlier reading used as a clue. **Planted anomaly:** a problem we insert on purpose so we know the right answer. **Bootstrap interval:** a margin of error from resampling whole buildings.